

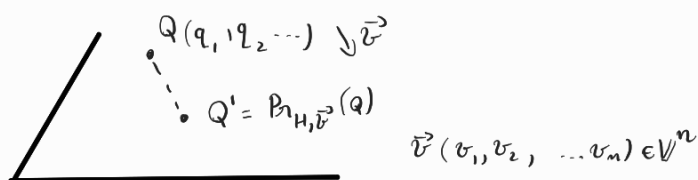
# (Parallel) Projections (& reflections) with respect to a hyperplane

Hyperplane: In an  $n$ -dim euclidian space, is a set of the form:

$$H = \{ (x_1, x_2, \dots, x_n) \in E^n \mid a_1 x_1 + a_2 x_2 + \dots + a_n x_n + a_{n+1} = 0 \}$$

If  $n=2 \Rightarrow H = \text{line}$

If  $n=3 \Rightarrow H = \text{plane}$



$$Q' = l \cap H \quad \text{where } l = Q + \langle \vec{v} \rangle : \begin{cases} x_1 = q_1 + t v_1 \\ x_2 = q_2 + t v_2 \\ \dots \\ x_n = q_n + t v_n \end{cases} \quad Q' = \begin{cases} x_1 = q_1 + t v_1 \\ \dots \\ x_n = q_n + t v_n \\ a_1 x_1 + \dots + a_n x_n + a_{n+1} = 0 \end{cases}$$

$$a_1 (q_1 + t v_1) + a_2 (q_2 + t v_2) + \dots + a_n (q_n + t v_n) + a_{n+1} = 0$$

$$\Rightarrow (a_1 q_1 + a_2 q_2 + \dots + a_n q_n) + t (a_1 v_1 + \dots + a_n v_n) + a_{n+1} = 0$$

$$t = - \frac{a_1 q_1 + a_2 q_2 + \dots + a_n q_n + a_{n+1}}{a_1 v_1 + a_2 v_2 + \dots + a_n v_n}$$

coordinates of  $Q'$ :

$$\begin{cases} x_1 = q_1 - \frac{a_1 q_1 + \dots}{a_1 v_1 + \dots} \cdot v_1 \\ \dots \\ x_n = q_n - \frac{a_1 q_1 + \dots + a_{n-1} q_{n-1}}{a_1 v_1 + \dots} \cdot v_n \end{cases}$$

Let  $p(x_1, \dots, x_n)$ ;  $p' = pr_{H, \vec{v}}(p) = (x'_1, \dots, x'_n)$

$$\Rightarrow \begin{pmatrix} x'_1 \\ x'_2 \\ \vdots \\ x'_n \end{pmatrix} = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix} - \frac{a_1 x_1 + \dots + a_n x_n + a_{n+1}}{a_1 v_1 + a_2 v_2 + \dots + a_n v_n} \begin{pmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{pmatrix}$$

## The tensor Product

$$\vec{v} (v_1, v_2, \dots, v_n), \vec{w} (w_1, \dots, w_n) \in V^n$$

$$\vec{v} \cdot \vec{w} = (v_1, v_2, \dots, v_n) \cdot \begin{pmatrix} w_1 \\ \vdots \\ w_n \end{pmatrix} = \vec{v} \cdot \vec{w}$$

$$\vec{v} \otimes \vec{w} = \begin{pmatrix} v_1 \\ \vdots \\ v_n \end{pmatrix} \cdot (w_1 \dots w_n) = \begin{pmatrix} v_1 w_1 & v_1 w_2 & \dots & v_1 w_n \\ \vdots & \vdots & & \vdots \\ v_n w_1 & v_n w_2 & \dots & v_n w_n \end{pmatrix}$$

↑  
tensor  
prod.

$$A, B \in M_n(\mathbb{R}) ; A = \begin{pmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ \vdots & \vdots & & \vdots \\ a_{n1} & a_{n2} & \dots & a_{nn} \end{pmatrix}$$

$$A \otimes B = \begin{pmatrix} a_{11}B & a_{12}B & \dots & a_{1n}B \\ \vdots & \vdots & & \vdots \\ a_{n1}B & a_{n2}B & \dots & a_{nn}B \end{pmatrix}$$

$$A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} \quad B = \begin{pmatrix} 0 & 1 \\ -1 & -1 \end{pmatrix}$$

$$A \otimes B = \begin{pmatrix} 0 & 1 & 0 & 2 \\ -1 & -1 & -2 & -2 \\ 0 & 3 & 0 & 4 \\ -3 & -3 & -4 & -4 \end{pmatrix}$$

Prop: The matrix form of the parallel projection is:  $P = \frac{\lim t}{a_1 x_1 + \dots + a_n x_n + a_{n+1}}$

$$\left[ P_{H, \vec{v}}(P) \right]_R = \frac{1}{(\lim t)(\vec{v})} (\vec{a} \vec{v} \text{Id}_n - \vec{v} \otimes \vec{a}) \cdot [P]_R - \frac{a_{n+1}}{(\lim t)\vec{v}} [\vec{v}]_R$$

$$\vec{a} = \vec{n}_H = (a_1, a_2, \dots, a_n)$$

$$\vec{v} = (v_1, \dots, v_n)$$

## Changing coordinates systems

$$\left. \begin{aligned} R &= (o, \overbrace{\vec{e}_1, \vec{e}_2, \dots, \vec{e}_n}^b) \\ R' &= (o', \underbrace{\vec{e}'_1, \vec{e}'_2, \dots, \vec{e}'_n}_{b'}) \end{aligned} \right\} \text{reference systems}$$

$$o, o' \in \mathbb{E}^n$$

$$\forall i, \quad \vec{e}_i, \vec{e}'_i \in V^n; \quad \forall P \in \mathbb{E}^n: [P]_R = [\vec{OP}]_b$$

The base-change matrix from  $R$  to  $R'$  is the base change matrix from  $b$  to  $b'$

$$M_{R',R} = M_{b',b} = [\text{id}]_{b,b'} = T_{b',b}$$

$$\forall P \in \mathbb{E}^n$$

$$[P]_{R'} = [\vec{O'P}]_{R'} = [\vec{O'P}]_{b'} = [\text{id}]_{bb'} [\vec{O'P}]_b$$

$$= M_{R',R} \underbrace{[\vec{O'P}]_b}_{\vec{OP} - \vec{OO'}} = M_{R',R} [O]_R = M_{R',R} \cdot [P]_R + \underbrace{M_{R',R} [\vec{O'O}]_b}_{[\vec{O'o}]_{b'}} = M_{R',R} \cdot [P]_R + [O]_{R'}$$

## Eigen values / vectors:

$$\text{Let } A = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

$$P_A(x) = \det(A - xI_2) = \begin{vmatrix} 1-x & 0 \\ 0 & -1-x \end{vmatrix} = (1-x)(-1-x) \Rightarrow \text{The eigen values are } \lambda_1=1; \lambda_2=-1$$

↑ egalezi cu 0  $\Rightarrow$  eigen value

$$V(\lambda_1) = \left\{ (x, y) \in \mathbb{R}^2 \mid A \begin{pmatrix} x \\ y \end{pmatrix} = \lambda_1 \begin{pmatrix} x \\ y \end{pmatrix} \right\}$$

$$= \left\{ (x, y) \in \mathbb{R}^2 \mid (A - \lambda_1 I_2) \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \right\} = \left\{ (x, y) \in \mathbb{R}^2 \mid \begin{pmatrix} 0 & 0 \\ 0 & -2 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \right\}$$
$$= \left\{ (x, y) \in \mathbb{R}^2 \mid -2y = 0 \right\} = \left\{ (x, 0) \mid x \in \mathbb{R} \right\}$$

$$= \langle (1, 0) \rangle$$

$$ma(\lambda_1) = 1$$

↑ algebraic multiplicity (de câte ori  $\lambda_1$  e root, ex: pt  $(x-1)^2 \Rightarrow x=1$   $ma(x)=2$ )

$$mg(\lambda_1) = \dim V(\lambda_1) = 1$$

M diagonalisable if  $ma(\lambda) = mg(\lambda), \forall \lambda \in \mathbb{R}$

## Axis and angle of rotation

$$O(n) = \{ A \in M_n(\mathbb{R}) \mid A \cdot A^t = I_n \}$$

$$SO(n) = \{ A \in O(n) \mid \det(A) = 1 \}$$

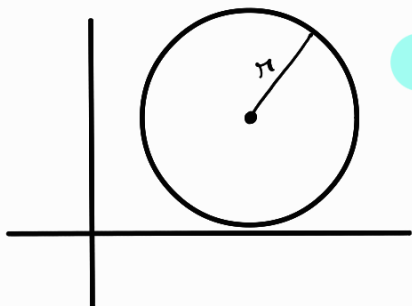
$$\text{If } A \in SO(n) \Rightarrow \text{Tr}(A) = 1 + 2\cos\theta; \theta - \text{angle of rotation}$$

$$\text{Fix}(A) = \left\{ (x, y, z) \mid A \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} x \\ y \\ z \end{pmatrix} \right\} - \text{axis of rotation}$$

Example:

$$\text{Fix}(A) = \left\{ \left( -\frac{\theta}{2}, 0, 1 \right) \mid \theta \in \mathbb{R} \right\} = \langle \left( -\frac{1}{2}, 0, 1 \right) \rangle$$

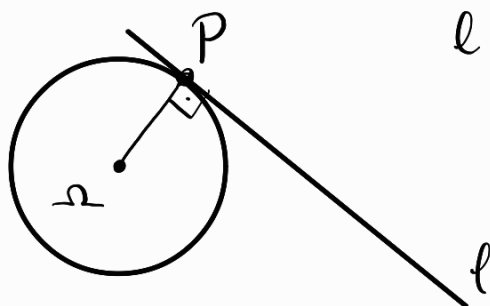
## Circle and ellipses



locus - set of points that have  
a certain property

$$C(\Omega, r) = (x - x_0)^2 + (y - y_0)^2 = r^2 \rightarrow \text{implicit eq.}$$

$$C(\Omega, r): \begin{cases} x = x_0 + r \cos t \\ y = y_0 + r \sin t \end{cases}, t \in [0, 2\pi)$$



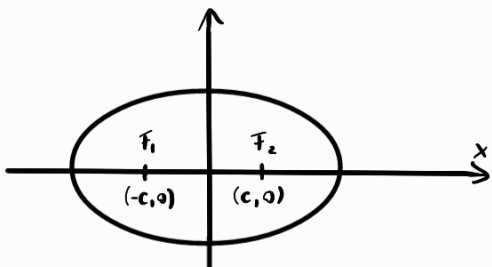
$l$  tangent to  $C(\Omega, r)$   
 $\Rightarrow \Omega P \perp l$

\* A point  $P(x_P, y_P)$  is on the line  $l: x - y - c = 0$  if  $x_P - y_P - c = 0$

$$d(A, B) = \sqrt{(x_A - x_B)^2 + (y_A - y_B)^2}; A(x_A, y_A) B(x_B, y_B)$$

$$d(A, l) = \frac{|a \cdot x_0 + b \cdot y_0 + c|}{\sqrt{a^2 + b^2}}; A(x_0, y_0) l: ax + by + c$$

# Ellipses:



Fix  $F_1, F_2$  two distinct points and  $a \in \mathbb{R}$

$\mathcal{E}$  = the locus of points  $P$  in the plane for which

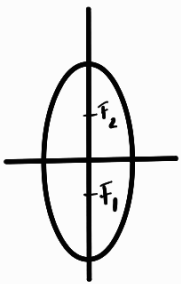
$$PF_1 + PF_2 = 2a$$

$$\mathcal{E} : \frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$

$$c = \sqrt{a^2 - b^2} \quad (a > b)$$

$$e = \frac{c}{a} \rightarrow \text{eccentricity}$$

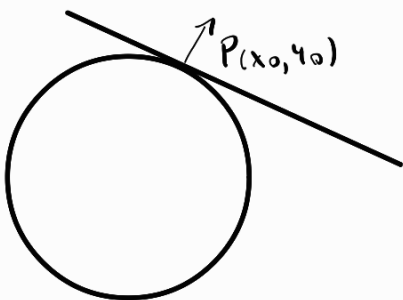
$F_1, F_2$  foci has to be on the longer axis



$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$

$$c = \sqrt{b^2 - a^2}$$

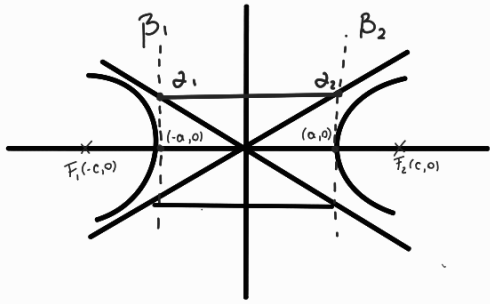
$$e = \frac{c}{b}$$



$$T_{\mathcal{E}(x_0, y_0)} : \frac{x \cdot x_0}{a^2} + \frac{y \cdot y_0}{b^2} = 1 \Leftrightarrow \frac{\partial F}{\partial x}(x_0, y_0)(x - x_0) + \frac{\partial F}{\partial y}(x_0, y_0)(y - y_0) = 0$$

$$F(x, y) = \frac{x^2}{a^2} + \frac{y^2}{b^2} - 1$$

# Hyperbolas and parabolas



Fix  $p_1, p_2$

H-locus of points  $M$  in the plane for which  $(MF_1 - MF_2) = 2a$

$F_1, F_2$  foci of the ellipse

$$H: \frac{x^2}{a^2} - \frac{y^2}{b^2} = 1 ; y = \pm \frac{b}{a} \sqrt{x^2 - a^2} ; c = \sqrt{a^2 + b^2} ; a, b > 0$$

H has 2 asymptotes at  $\pm \infty$  :

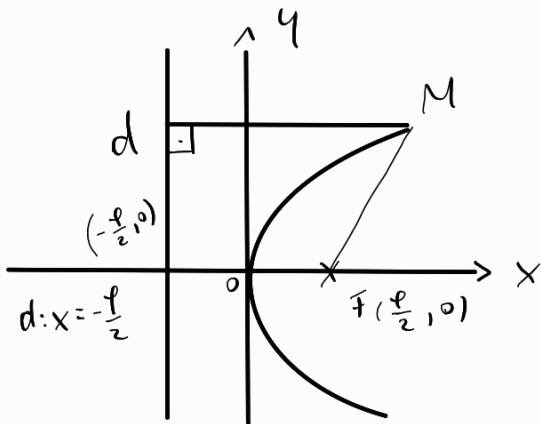
- $\alpha_1: y = -\frac{b}{a}x$
- $\alpha_2: y = \frac{b}{a}x$

If we trace the lines  $\alpha_1, \alpha_2$  along with  $\beta_1 = x = -a ; \beta_2 = x = a$   
we obtain a rectangle with sides  $2a$  and  $2b$

$$T_H(x_0, y_0): \frac{x x_0}{a^2} - \frac{y y_0}{b^2} = 1 \Leftrightarrow \frac{\partial F}{\partial x}(x_0, y_0)(x - x_0) + \frac{\partial F}{\partial y}(x_0, y_0)(y - y_0) = 0$$

$$F(x, y) = \frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$$

## Parabolas



$F$  point (the focus)

$d$  line (the directrix)

$P$  = locus of points in the plane  
equidistant to  $F$  and  $d$

$$MF = \text{dist}(M, d)$$

$$P: y^2 = 2px \Leftrightarrow x = \frac{y^2}{2p} \Leftrightarrow y = \pm \sqrt{2px}$$

$$T_P(x_0, y_0): y y_0 = p(x + x_0) \Leftrightarrow \frac{\partial F}{\partial x}(x_0, y_0)(x - x_0) + \frac{\partial F}{\partial y}(x_0, y_0)(y - y_0) = 0$$

$$F(x, y) = y^2 - 2px$$

# Canonical form:

$$Q: a_{11}x^2 + 2a_{12}xy + a_{22}y^2 + a_{10}x + a_{01}y + a_{00} = 0$$

$$M(Q) = \begin{pmatrix} a_{11} & a_{12} \\ a_{12} & a_{22} \end{pmatrix}$$

→ Find the eigen values (root of  $P_{M(Q)}(x) = \begin{vmatrix} a_{11}-x & a_{12} \\ a_{12} & a_{22}-x \end{vmatrix}$ )

→ Find the eigen vectors and from them select a right oriented orthonormal basis (B)

Step I:

$$B = (\vec{v}_1, \vec{v}_2); E = (\vec{e}_1, \vec{e}_2) \text{ initial basis} \quad (AB)^t = B^t A^t$$

$$M_{E,B} = [id]_{B,E} = ([\vec{v}_1]_E, [\vec{v}_2]_E)$$

$$\text{We have } \begin{pmatrix} x \\ y \end{pmatrix} = M_{E,B} \begin{pmatrix} x_1 \\ y_1 \end{pmatrix}$$

coord in the old basis E

coord in the new basis B

$$Q: (x \ y) \begin{pmatrix} a_{11} & a_{12} \\ a_{12} & a_{22} \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} + (a_{10} \ a_{01}) \begin{pmatrix} x \\ y \end{pmatrix} + a_{00} = 0$$

$$(x \ y) = \begin{pmatrix} x \\ y \end{pmatrix}^t = \begin{pmatrix} x_1 \\ y_1 \end{pmatrix}^t \cdot M_{E,B}^t = (x_1 \ y_1) M_{E,B}$$

$$Q: (x_1 \ y_1) M_{E,B}^t M_{(Q)} M_{E,B} \begin{pmatrix} x_1 \\ y_1 \end{pmatrix} + (a_{10} \ a_{01}) M_{E,B} \begin{pmatrix} x_1 \\ y_1 \end{pmatrix} M_{E,B}^t M_{(Q)} M_{E,B} = \begin{pmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{pmatrix} \quad \lambda_1, \lambda_2 - \text{eigenvalues of } M_{(Q)}$$

$$Q: \lambda_1 x_1^2 + \lambda_2 y_2^2 + 2\alpha x_1 + \beta y_1 + a_{00} = 0$$

Step II force squares ( $\lambda_1, \lambda_2 \neq 0$ )

ELLIPSES  
HYPERBOLA

$$Q: \lambda_1 \left( x_1^2 + 2 \frac{\alpha}{2\lambda_1} x_1 + \frac{\alpha^2}{4\lambda_1^2} \right) + \lambda_2 \left( y_1^2 + 2 \frac{\beta}{2\lambda_2} y_1 + \frac{\beta^2}{4\lambda_2^2} \right) + a_{00} - \frac{\alpha^2}{4\lambda_1} - \frac{\beta^2}{4\lambda_2}$$

← K

$$Q: \lambda_1 \left( x_1 + \frac{\alpha}{2\lambda_1} \right)^2 + \lambda_2 \left( y_1 + \frac{\beta}{2\lambda_2} \right)^2 + K = 0$$

$$\begin{cases} x_2 = x_1 + \frac{\alpha}{2\lambda_1} \\ y_2 = y_1 + \frac{\beta}{2\lambda_2} \end{cases} \Rightarrow Q: \lambda_1 x_2^2 + \lambda_2 y_2^2 + K = 0$$